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Upgrading to React 17 and Webpack 5

How I upgraded my project to React 17 and Webpack 5





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a **No New Feature** release.

I was a bit disappointed initially but anyway went ahead with upgrading to React 17 and to Webpack 5.

Later, I noticed several enhancements under the hood with React 17, which I strongly recommend updating.

This article will discuss the steps to upgrading a React project to React 17 and Webpack 5 with my own set of tips to avoid common pitfalls.

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Why Should You Upgrade?

Generally, as developers, we think twice before upgrading a project to the latest versions because there could be breaking changes, dependency issues, depreciation warnings, etc. Therefore, let's get a basic understanding of the impact of upgrading to React 17 and Webpack 5.

React 17

Although there are no significant features in this version, it sets the foundation for **gradual React upgrades** for the future.

Therefore, when you upgrade from React 16 to 17, you must upgrade the whole project at once.

However, from React 17 onwards, there will be two upgrade options. Hereafter, you can decide to upgrade the entire project at once or in parts to the latest version. Most importantly, you don't need to worry about breaking changes.

Gradual updates or quite simply, being able to use more than a single version of React

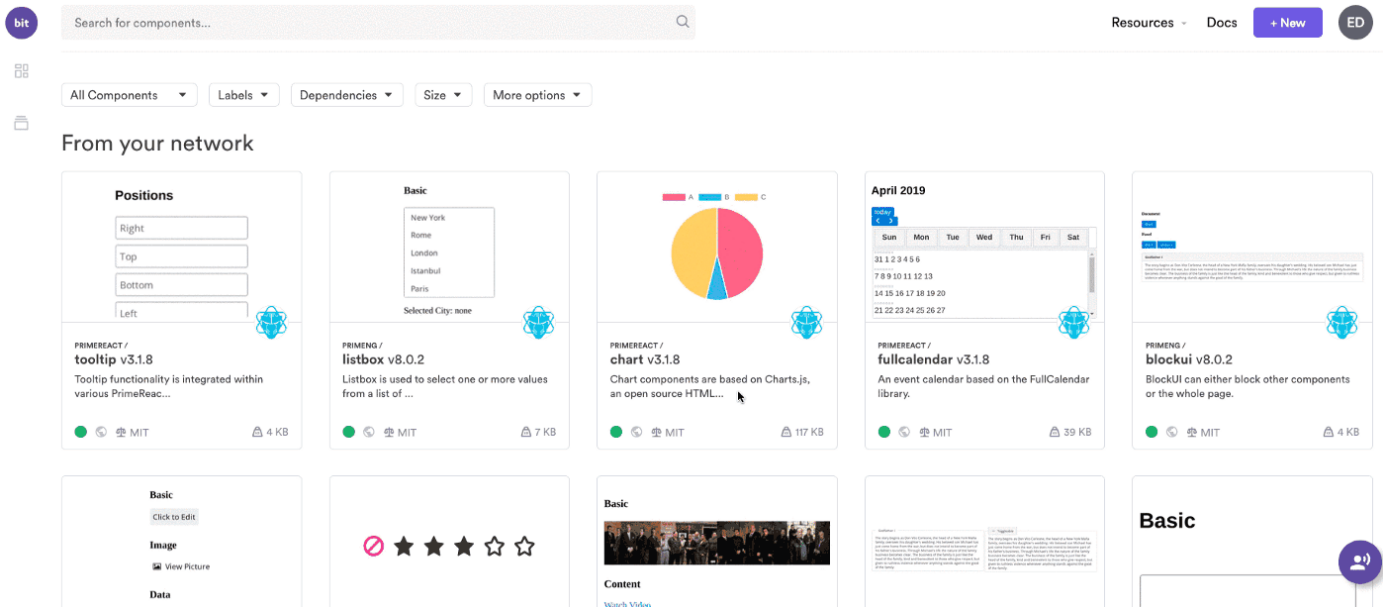




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these “components” are full-pages or widgets that make up an independent micro-frontend.



React components shared on [Bit.dev](https://bit.dev)

Besides, there are many other changes in React 17 like New JSX Transform, Effect Cleanup Timing, Native Component Stacks, and you can find more about those in their [documentation](#).

Webpack 5

Almost after 2 years of silence, the Webpack team have released their latest version as Webpack 5. The good news is that you can find many exciting features that mainly focuses on;

- Improving build time with persistent caching.
- Reducing bundle size.
- Long term caching.

Also Webpack 5 comes with some breaking changes



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How to Upgrade

I will show you the upgrading steps to React 17 and Webpack 5, using a React 16 project that uses TypeScript.

React 17

You can use either NPM or Yarn to do the upgrades, and it's just a single line command. I will upgrade the React version and react-dom version to React 17 and react-scripts to version 4 using the following command.

```
+ react-dom@17.0.0
+ react-scripts@4.0.0
+ react@17.0.0
added 4 packages from 1 contributor, removed 3 packages, updated 10 packages and audited 2127 packages in 84.601s

6 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Screenshot by Author

NPM:

```
npm install react-scripts@4.0.0 react@17.0.0 react-dom@17.0.0
```

Yarn:

```
yarn upgrade react-scripts@4.0.0 react@17.0.0 react-dom@17.0.0
```

Webpack 5

Similarly, you can upgrade the Webpack version of your project to the latest using Yarn or





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```
+ webpack@5.0.0-rc.6
added 13 packages from 41 contributors, updated 6 packages, moved 23 packages and audited 2140 packages in 18.389s

121 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

617 | 3

Screenshot by Author

NPM:

```
npm install webpack@next
```

Yarn:

```
yarn add webpack@next
```

That's it. Now you have upgraded your project into React 17 and Webpack 5.

But the upgrade isn't over yet. It would be best to make several changes to your project to get the best out of this release. Also, there are places where this upgrading process isn't that smooth. So, let's explore where things could go wrong and solve them in the following section.

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Common Issues You May Face After Upgrading

When I upgraded my React 16 project to React 17 and Webpack 5, I faced a few issues and had to find ways to fix these.

Using new JSX transform

With this new JSX transform, you will no longer need to import React on each component.





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NPM:

```
npm install @babel/core@^7.9.0 @babel/preset-react@^7.9.0 --dev
```

Yarn:

```
yarn add @babel/core@^7.9.0 @babel/preset-react@^7.9.0 --dev
```

Then you need to make a small change in your babel configuration file as follows:

```
{
  "presets": [
    ["@babel/preset-react", {
      "runtime": "automatic"
    }]
  ]
}
```

Now, you are good to clean up all import React statements in your project.

Fixing react-scripts & Webpack compatibility issues

When I run the application after upgrading, I received a runtime error saying that there is a mismatch between Webpack versions.

```
The react-scripts package provided by Create React App requires a dependency:
  "webpack": "4.44.2"

Don't try to install it manually: your package manager does it automatically.
However, a different version of webpack was detected higher up in the tree:

D:\Projects\popomates_web\node_modules\webpack (version: 5.0.0-rc.6)
```

The error message appeared after the upgrade





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1. Delete `package-lock.json` and `node_modules` in your project folder.
2. Remove `webpack` from `dependencies` and `devDependencies` in the `package.json`.
3. Run `npm install`.

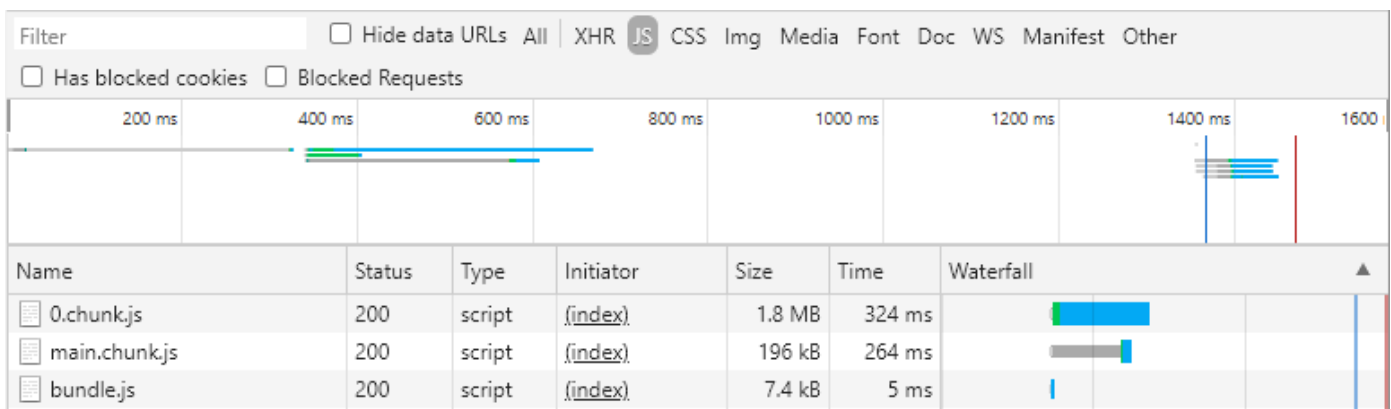
If you still get an error regarding react-scripts compatibility, update your `tsconfig.json` as follows:

```
{
  "compilerOptions": {
    "noFallthroughCasesInSwitch": true,
    ...
  },
  ...
}
```

...

Final Thoughts

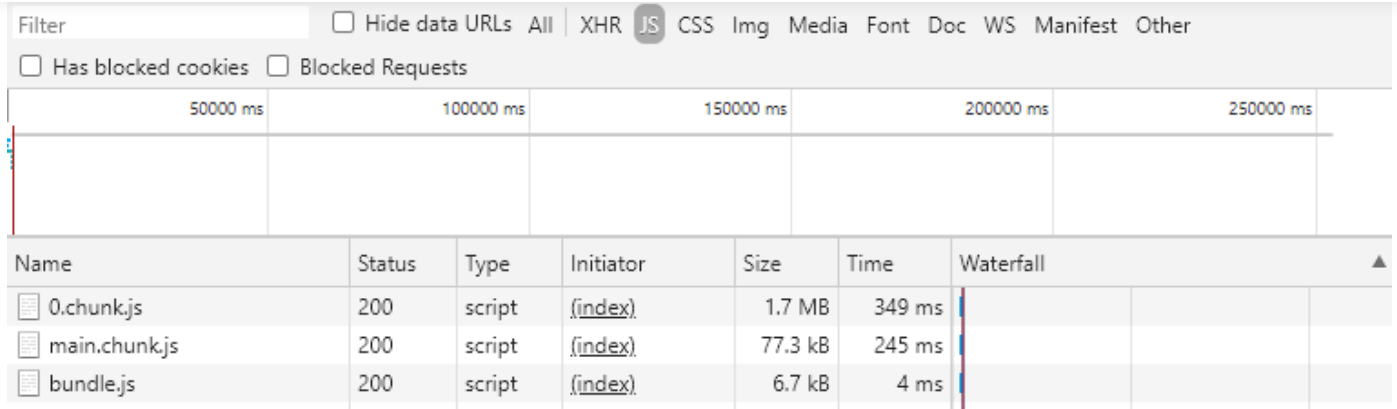
When I ran my project for the first time after the upgrade, I didn't feel much difference in its performance. Since Webpack 5 comes with features to reduce bundle size, I analyzed my application's size and the load time before and after the upgrade.





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React 17 + Webpack 5

As you can see, there are small changes in both file sizes and time, and these differences can get higher with the size of your project.

However, if you decide to upgrade your project, the problems you will encounter could differ from mine, based on your project dependencies, the version you are upgrading from &, etc. Therefore keep some room to fix any unknown issues that could appear.

Therefore it's important to keep some room to fix any unknown issues after doing the upgrade.

Therefore, I think upgrading projects to the latest versions will reduce the hassle of waiting for the next versions. It is because both React 17 and Webpack 5 sets a foundation for upcoming features.

Both React 17 and Webpack 5 contain a minimum number of breaking changes. And having React 17 ensures a gradual upgrade process from React 18 onwards. Therefore this is the best time to make some investment for the future by upgrading your projects.

Thank you for Reading!!!

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